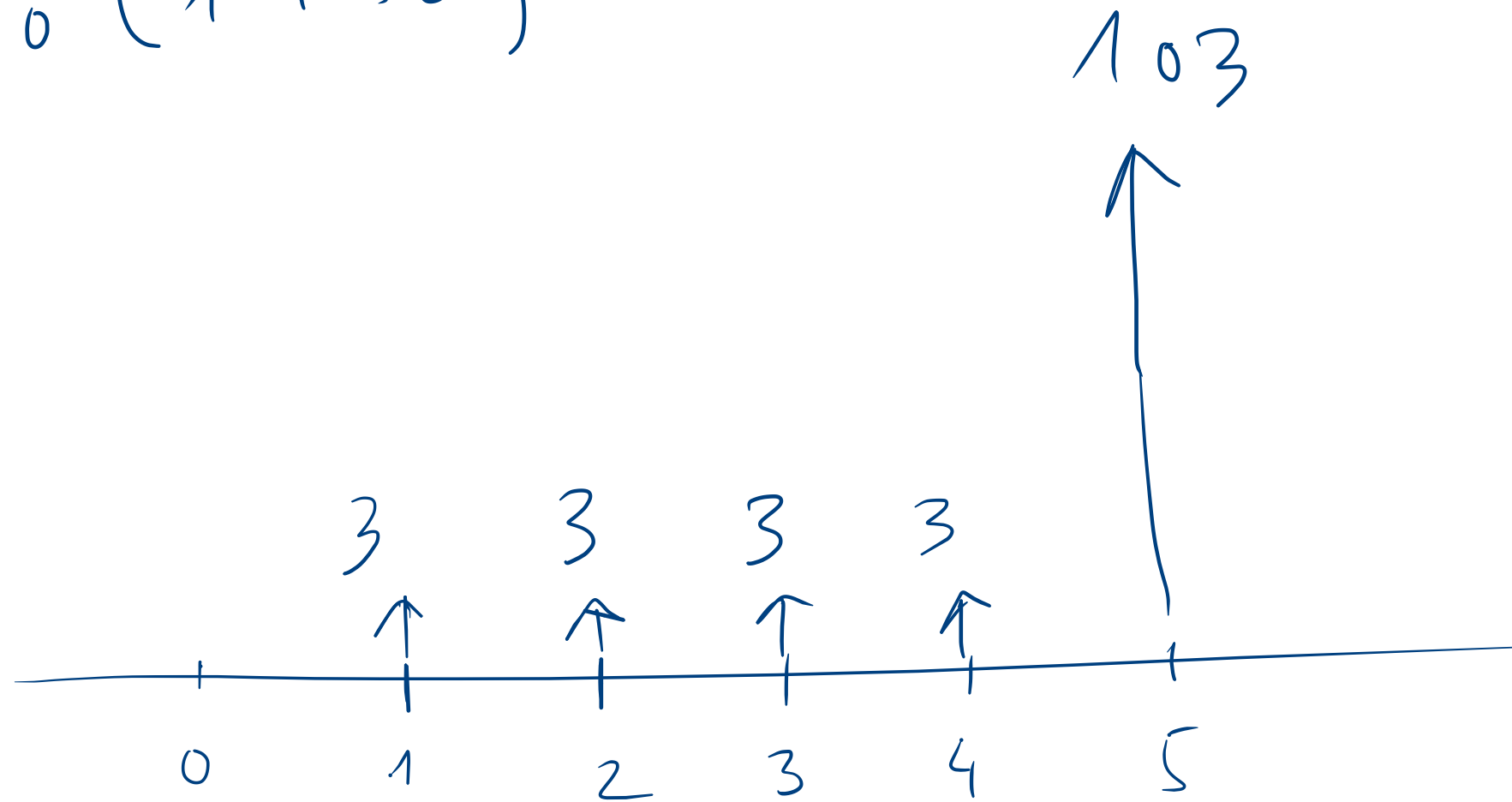


$$V_n = V_0 (1 + r)^n$$



Arbitrage : portefeuille X

$$\text{t.g.} \quad X_0 = 0 \quad X_T \geq 0 \quad P(X_T > 0) > 0$$

Preemptive vs Postemptive

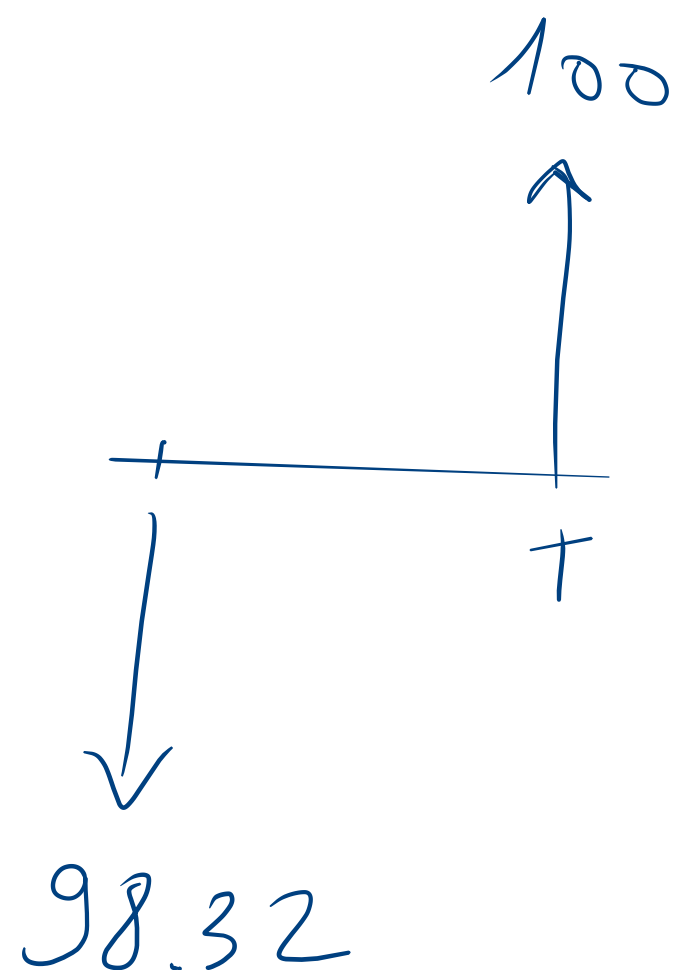
$$\text{POST: } 1 \xrightarrow{1\text{an}} 1+n$$

$$\text{PRE: } 1-n' \rightarrow 1$$

$$\left\{ \begin{array}{l} \frac{V_f}{V_i} = \frac{1+n}{1} = \frac{1}{1-n'} \end{array} \right.$$

$$n' = \frac{n}{1+n}$$

$$n' \times 10\text{€} = \frac{n \times 10\text{€}}{1+n}$$



$$V_h = V_0 (1 + r)^h$$

$$C \times \left(1 + \frac{10\%}{2}\right)^2$$

$$C \times (1 + 10\% \times 0,5)^2$$

$$\frac{dS_t}{S_t} = r dt + \sigma dW_t$$

$$dV_t = rV_t dt$$

$$1 + r_a = e^{r_c}$$

$$r_c = \ln(1 + r_a)$$

Yield To Maturity

$$DF(T) = \frac{1}{(1 + r(T))^T} \text{ ou } e^{-r(T)T}$$

$T \mapsto r(T)$ Courbe ZC

$$\rightarrow r(T) = \left(\frac{1}{DF(T)} \right)^{1/T} - 1$$